

Logan Mathew Walsh

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Education

University of Auckland

Bachelor of Science, Majoring in Computer Science.

March 2023 – November 2025

Auckland, New Zealand

- GPA: 7.9/9
- Major Awards: First in Course (COMPSCI 101), First in Course (COMPSCI 130)

Experience

Associate Software Developer

Current Limited

August 2026 - Current

Auckland, New Zealand

Full Stack Developer Intern

Promptech

July 2025 – November 2025

Auckland, New Zealand

- Collaborated within a six-member team using Agile methodologies to design, develop, and deploy an AWS-hosted GenAI playground platform for educational and workshop use.
- Developed and integrated PostgreSQL databases and FastAPI service layers with React Typescript components, enabling efficient data flow and responsive UI updates.
- Built and maintained secure RESTful APIs supporting the frontend, backend and wallet payment features.
- Integrated Stripe's payment gateway for top-ups, implemented balance tracking and per-request credit deduction.
- Contributed across the full stack, enhancing user experience, data reliability, and overall system performance.

Teaching Assistant

The University of Auckland

July 2024 – November 2025

Auckland, New Zealand

- Assisted weekly lab sessions for 40+ students in COMPSCI 130 (Software Fundamentals), teaching object-oriented programming, data structures, and algorithms.
- Provided personalized guidance to students by clarifying concepts, answering questions, and assisting with complex problems to strengthen problem-solving skills and practical coding application.

Software Specialist Intern

Business Continuum

January 2024 – February 2024

Melbourne, Australia

- Consulted with clients to discuss database refactoring, management, and financial system optimization.
- Led a project from discovery to delivery, planning process design workshops, client reviews, and training sessions.
- Provided live support and training, both on-site and remote to ensure customer success and satisfaction.

Projects

RISC-V Virtual Machine | C, React.js, Electron.js, Node.js

- Built a RISC-V Virtual Machine using the RV32I base instruction set, implementing 33 instructions with full memory management, custom dynamic allocation using heap banks, and memory-mapped I/O virtual routines.
- Created a custom heap allocator that manages 128 memory banks with support for malloc/free operations, including boundary checking, consecutive bank allocation, and memory safety guarantees.
- Developed a simple Electron-based GUI debugger with React frontend, providing real-time VM execution, I/O communications, register visualisation and automatic binary compilation.

Stock Analysis Platform | Python FastAPI, TypeScript, React, OpenAI

- Developed an analytical tool that helps users make better investment decisions.
- Provides live data, expert opinions, recent and relevant news articles, AI generated summaries and suggestions, trends, and in the future customisable ML algorithms to help predicts trends.

Technical Skills

Languages: C/C++, Python, Rust, TypeScript

Frameworks: FastAPI, PyTest, React, Next.js, Node.js, Electron.js

Databases: PostgreSQL, SQLite, Redis

Tools: Unix/Linux, Git, AWS (EC2, Cognito, Bedrock), Docker, CI/CD, Postman